

RECONCILIATION AUDIT
REPORT

Evidence-backed verification of the active land dataset

STATUS: REVIEW REQUIRED

AREA: Delhi NCR — Urban District

1. DATASET INFORMATION

Dataset Name:	Delhi NCR — Urban District	Dataset ID:	delhi-urban
City / Urban Area:	Delhi NCR (Urban District)	Dataset Version:	v1.2
Processing Date:	2026-09-03	Authoritative CRS:	EPSG:32643 (UTM 43N) → EPSG:4326
Total Entities:	72 Parcels	Verification State:	REVIEW REQUIRED

2. RECONCILIATION SUMMARY

Metric Category	Parcel Count	Distribution (%)	Audit Status
Verified (Auto-Harmonized)	44	61.1%	VERIFIED · READY
Surveyor Review Required	22	30.6%	REVIEW RECOMMENDED
Spatial Conflict Detected	6	8.3%	CONFLICT QUEUED
Total Land Parcels Analyzed	72	100.0%	Avg Confidence: 88.4%

3. SOURCE EVIDENCE

Authoritative Data Source	Layer Type	Participating Parcels	Coverage	Status
Revenue Cadastral (Khasra/FMB)	CADASTRAL	72 parcels	100.0%	VERIFIED
Municipal Property GIS	MUNICIPAL	72 parcels	100.0%	VERIFIED
Drone Orthomosaic (ORI)	ORI	72 parcels	100.0%	VERIFIED
AI Footprint Extraction	AI	72 parcels	100.0%	VERIFIED
Survey of India CORS RTK Benchmarks	RTK	8 parcels	11.1%	VERIFIED

4. SPATIAL QUALITY BENCHMARKS

Quality Measure	Measured Value	Authoritative Threshold	Audit Status	Evaluation Purpose
IoU	89.6% (0.896)	≥ 85.0%	PASSED	Measures how closely source boundaries overlap with the reconciled boundary.
Boundary Agreement	88.3% (0.883)	≥ 80.0%	PASSED	Measures consistency between available source boundaries.

Ground Truth Agreement	87.8% (0.878)	≥ 80.0%	PASSED	Measures agreement with verified survey or ground-reference data.
Verification Confidence	88.4% (0.884)	≥ 85.0%	PASSED	Shows the current confidence assigned to the reconciliation result.

5. SPATIAL CONFLICT SUMMARY

CONFLICT CLASSIFICATION	INCIDENCE COUNT	DISCREPANCY DESCRIPTION	RESOLUTION STATUS
Infrastructure Crossing	9	Proximity or spatial overlap with designated road ROW or SWD buffer	PENDING REVIEW
Source Mismatch	9	Area or geometry shape disagreement between revenue deeds and municipal GIS	PENDING REVIEW
Boundary Offset	6	Lateral coordinate discrepancy between cadastral boundary and building edge	PENDING REVIEW
Geometry Issue	3	Self-intersection, sliver polygon, or vertex collinearity anomaly	PENDING REVIEW
Centroid Drift	1	Centroid displacement exceeding local authoritative tolerance threshold	PENDING REVIEW

6. PARCEL RECONCILIATION LEDGER (ISSUES REQUIRING INSPECTION)

The following parcels exhibit coordinate boundary drift, lower consensus overlap, or multi-source discrepancies requiring cadastral surveyor review:

PARCEL ID	SURVEY NO	CONFLICT TYPE	SOURCES	DRIFT	IOU	CONF	STATUS
DEL-102	DL-NDMC-118/B	Boundary Offset	3 Sources	1.62m	0.82	80%	REVIEW
DEL-106	DL-NDMC-186/F	Geometry Issue	3 Sources	4.0m	0.75	66%	CONFLICT
DEL-107	DL-NDMC-203/A	Infrastructure Crossing	3 Sources	1.54m	0.81	79%	REVIEW
DEL-109	DL-NDMC-237/C	Boundary Offset	3 Sources	0.9m	0.85	81%	REVIEW
DEL-112	DL-NDMC-288/F	Source Mismatch	3 Sources	1.06m	0.85	84%	REVIEW
DEL-115	DL-NDMC-339/C	Source Mismatch	3 Sources	1.62m	0.86	81%	REVIEW
DEL-117	DL-NDMC-373/E	Source Mismatch	3 Sources	1.22m	0.83	83%	REVIEW
DEL-118	DL-NDMC-390/F	Boundary Offset	3 Sources	2.5m	0.64	62%	CONFLICT
DEL-122	DL-NDMC-458/D	Infrastructure Crossing	3 Sources	0.98m	0.84	82%	REVIEW
DEL-123	DL-NDMC-475/E	Source Mismatch	3 Sources	1.22m	0.83	83%	REVIEW
DEL-127	DL-NDMC-143/C	Infrastructure Crossing	3 Sources	1.46m	0.86	81%	REVIEW
DEL-130	DL-NDMC-194/F	Geometry Issue	3 Sources	4.0m	0.67	66%	CONFLICT
DEL-132	DL-NDMC-228/B	Boundary Offset	3 Sources	1.14m	0.82	80%	REVIEW
DEL-134	DL-NDMC-262/D	Source Mismatch	3 Sources	1.62m	0.84	82%	REVIEW
DEL-137	DL-NDMC-313/A	Source Mismatch	3 Sources	1.46m	0.82	79%	REVIEW

7. EVIDENCE COVERAGE & AUDIT PILLARS

EVIDENCE DOMAIN	COVERAGE STATE	PROVENANCE & METHODOLOGY NOTE
Geometry Validation	COMPLETE	All polygon boundaries checked for closure, orientation, and valid topology
CRS Validation	COMPLETE	Source geometries transformed to EPSG:32643 (UTM 43N) → EPSG:4326 with zero angular distortion
Source Evidence	COMPLETE	5 authoritative source layers participating with immutable provenance
Ground Truth	COMPLETE	8 CORS / RTK GNSS benchmark survey points correlated
Historical Evidence	NOT AVAILABLE	Prior year cadastral revision snapshots attached for temporal comparison
Infrastructure Evidence	COMPLETE	Road Right-of-Way and Stormwater Drainage corridor crossing evaluation active

8. RECONCILIATION PIPELINE PROGRESSION

STAGE	PIPELINE OPERATIONAL STAGE	PROCESSING STATE	STAGE DESCRIPTION
01	INGEST (01 Multi-Source Ingestion)	COMPLETED	Cadastral, Municipal & Drone ORI package validation
02	AI_02B (02B Mask R-CNN Inference)	COMPLETED	Drone ORI footprint instance segmentation (ResNet-50-FPN)
03	AI_02C (02C Polygon + CRS)	COMPLETED	Contour vectorization & metric affine georeferencing
04	STAGE_03 (03 Schema Normalization)	COMPLETED	Triple-source convergence (Cadastral, Municipal, AI Source)
05	MATCH (07-08 STRtree Matching)	COMPLETED	R-tree candidate filtering & multi-signal matching
06	COMPARE (09 IoU & Drift Evaluation)	COMPLETED	IoU, centroid drift, and area delta comparison
07	VERIFY (10-12 Consensus & Review)	REVIEW	Ground truth and policy threshold verification
08	REGISTER (13-16 Authoritative Register)	NOT STARTED	Final legal cadastre registry commit

9. AUDIT HISTORY & LIFECYCLE EVENTS

EVENT TIMESTAMP	ACTION / LIFECYCLE EVENT	ACTOR / SUBSYSTEM	STATUS
2026-09-03 10:14:22	Dataset package imported and validated	Data Ingestion Engine	COMPLETED
2026-09-03 10:15:08	Spatial CRS projected to metric UTM and normalized	CRS Safety Engine	COMPLETED
2026-09-03 10:16:45	8-signal multi-source candidate spatial indexing completed	STRtree Matcher	COMPLETED
2026-09-03 10:18:12	Autonomous multi-source boundary consensus evaluated	Consensus Engine	COMPLETED
2026-09-03 10:19:30	28 parcels queued for surveyor inspection	State Machine	REVIEW REQUIRED
2026-09-03 10:21:04	Survey of India CORS RTK ground-truth checkpoints attached	GNSS Verifier	COMPLETED

10. CURRENT DATASET STATUS & SPATIAL VERIFICATION STATEMENT

The active dataset **Delhi NCR — Urban District** (Delhi NCR — Urban District) currently holds an audit status of **REVIEW REQUIRED**. A total of 44 of 72 parcels (61.1%) meet the strict dual-threshold criteria ($\text{IoU} \geq 0.85$ and Centroid Drift < 2.0 meters) and are certified for authoritative land record workflows. The remaining 28 parcels have been formally queued in the Conflict Ledger for surveyor inspection.

STATUTORY NOTICE: This document provides spatial geometry reconciliation, multi-source sensor agreement, and technical evidentiary verification only. It does not constitute a final judicial determination of legal title or ownership. Administrative registration of final boundaries must follow statutory procedures stipulated under the relevant State Land Revenue Act.

11. DATA SOURCES & RECONCILIATION METHODOLOGY

Methodological Architecture: Spatial reconciliation in TERRANODE employs an 8-signal scoring matrix (combining Intersection-over-Union, centroid offset, area delta, perimeter agreement, shape similarity, and bounding-box overlap) indexed with Shapely STRtrees. Coordinate references are validated in native projected UTM metric systems prior to comparison, eliminating angular degree distortion. Ground-truth verification uses Survey of India CORS GNSS stations as authoritative reference coordinates.